

CSE260

Assignment - 04

Abrar Fahim

ID: 22201689

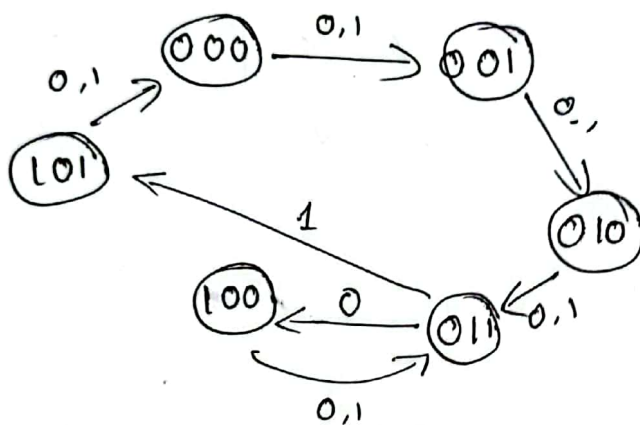
Sec: 02

Date: 4/5/25

8) let,

CSE110 - 000
 CSE111 - 001
 CSE220 - 010
 CSE221 - 011
 CSE331 - 100
 CSE321 - 101

T excitation		
Q	Q ⁺	T
0	0	0
0	1	1
1	0	1
1	1	0



	A	B	C	X	A ⁺	B ⁺	C ⁺	T _A	T _B	T _C
0	0	0	0	0	0	0	1	0	0	1
1	0	0	0	1	0	0	1	0	0	1
2	0	0	1	0	0	1	0	0	1	1
3	0	0	1	1	0	1	0	0	1	1
4	0	1	0	0	0	1	1	0	0	1
5	0	1	0	1	0	1	1	0	0	1
6	0	1	1	0	1	0	0	1	1	1
7	0	1	1	1	1	0	1	1	0	0

	A	B	C	X	A ⁺	B ⁺	C ⁺	T _A	T _B	T _C
8	1	0	0	0	0	1	1	1	1	1
9	1	0	0	1	0	1	1	1	1	1
10	1	0	1	0	0	0	0	1	0	1
11	1	0	1	1	0	0	0	1	0	1
12	1	1	0	0	x	x	x	x	x	x
13	1	1	0	1	x	x	x	x	x	x
14	1	1	1	0	x	x	x	x	x	x
15	1	1	1	1	x	x	x	x	x	x

AB CX

	C'x'	C'x	Cx	Cx'
A'B'			1	
A'B			1	1
AB	x	x	x	x
AB'	1	1	1	1

$$T_A = A + B + C$$

AD CX

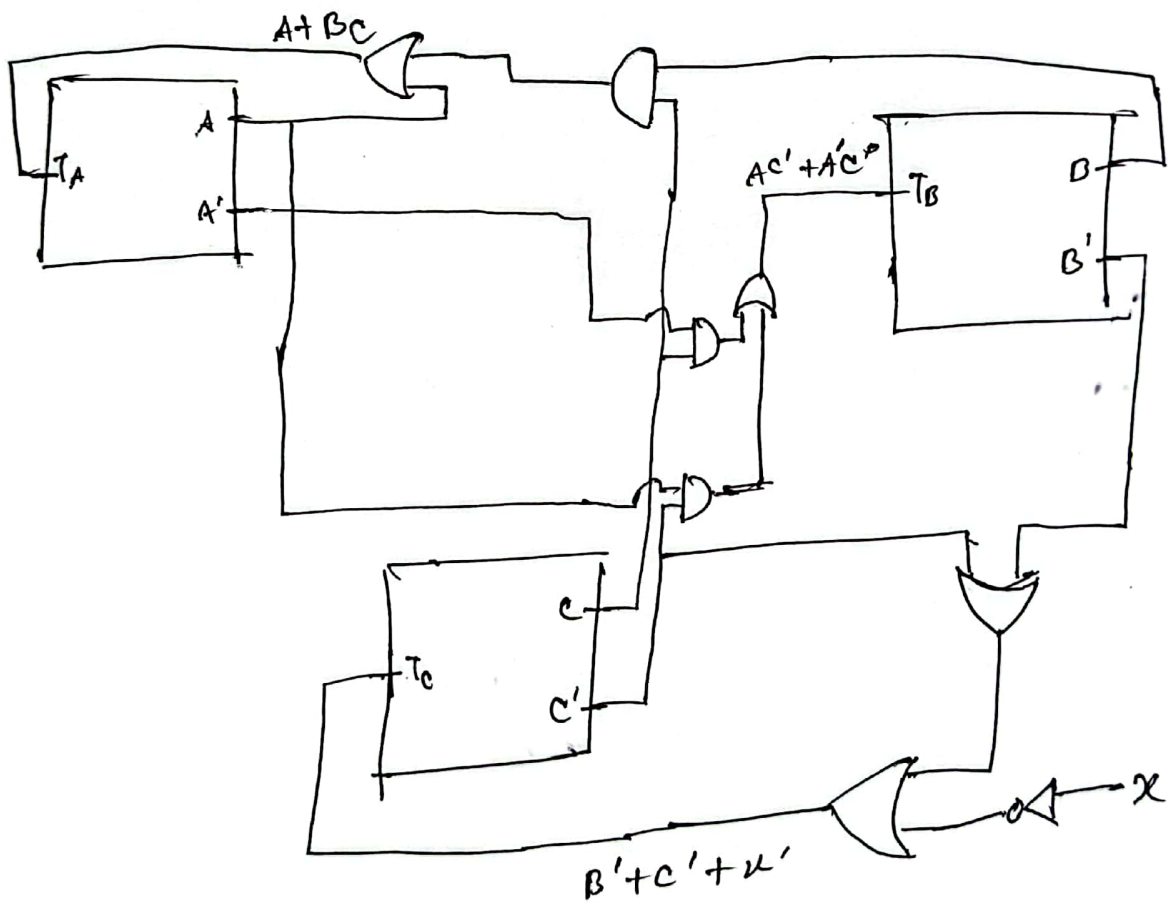
	C'x'	C'x	Cx	Cx'
A'B'			1	1
A'D			1	1
AB	x	x		
AB'	1	1		

$$T_B = AC' + A'C$$

$\backslash c$	c'	c	c'	c	c'
AB					
$A'B'$	1	1	1	1	1
$A'B$	1	1			1
AB	x	x	x	x	x
AB'	1	1	1	1	1

$$T_c = B' + C' + x'$$

~~Handwritten scribble~~



Circuit using T-flip flops.